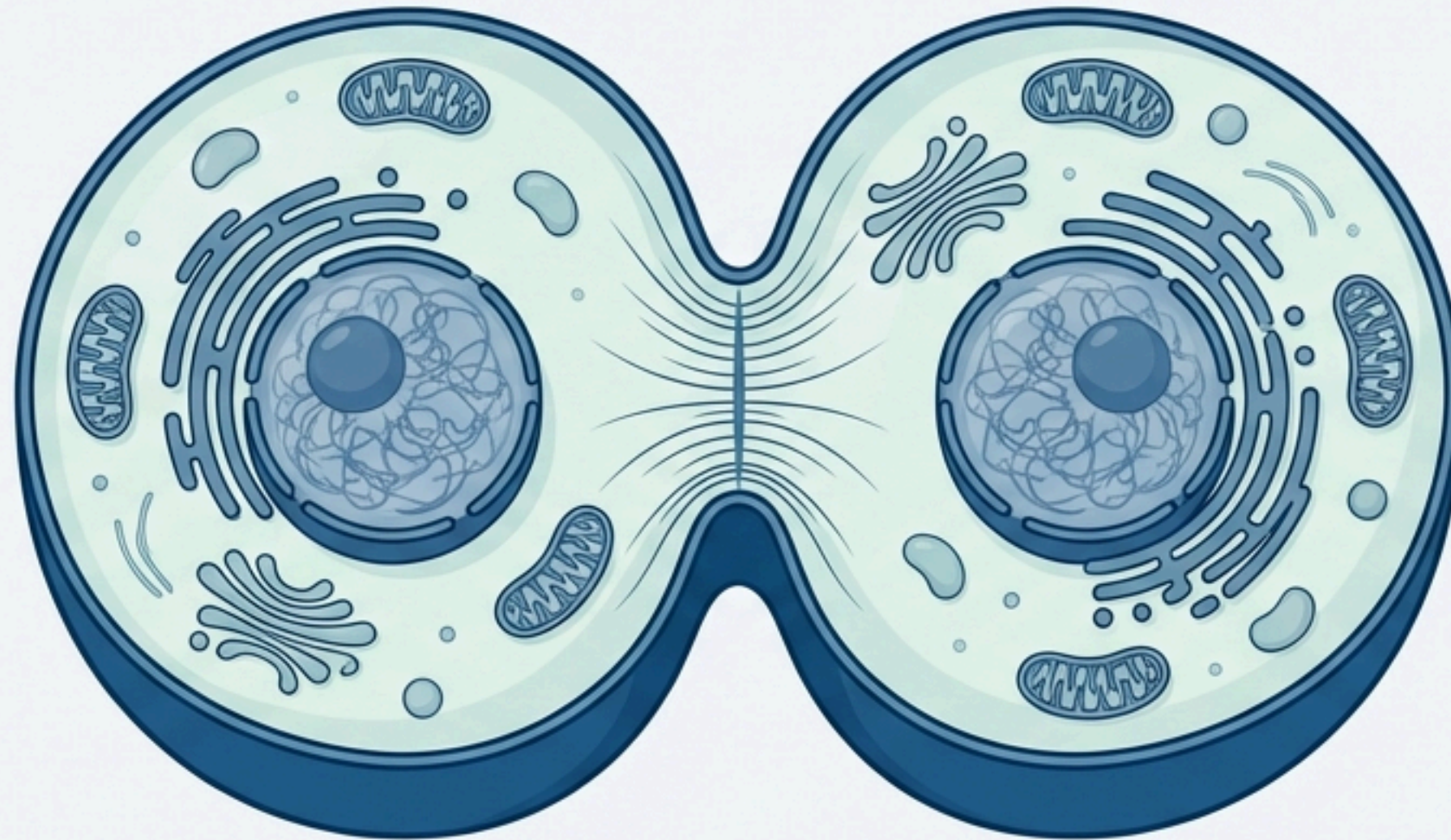


THE ARCHITECTURE OF CELLULAR REPRODUCTION

A STRATEGIC REVIEW OF CELL CYCLE & MITOSIS FOR NEET-UG

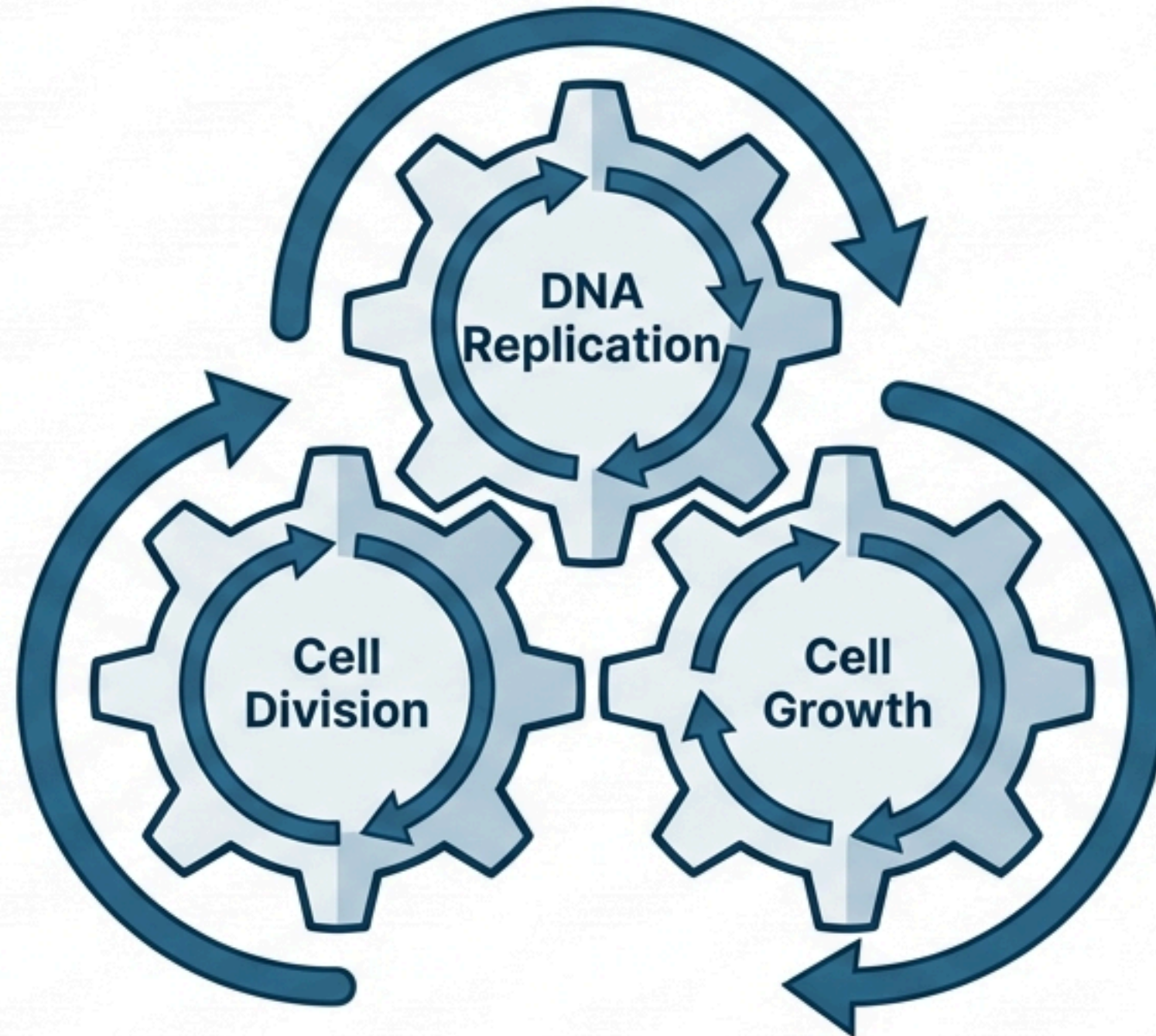


All organisms, from the smallest to the largest, begin life as a single cell.
This deck decodes the genetic control and physical mechanics of how
that single cell divides to form millions.

The Coordinated Trinity

Division, Replication, and Growth

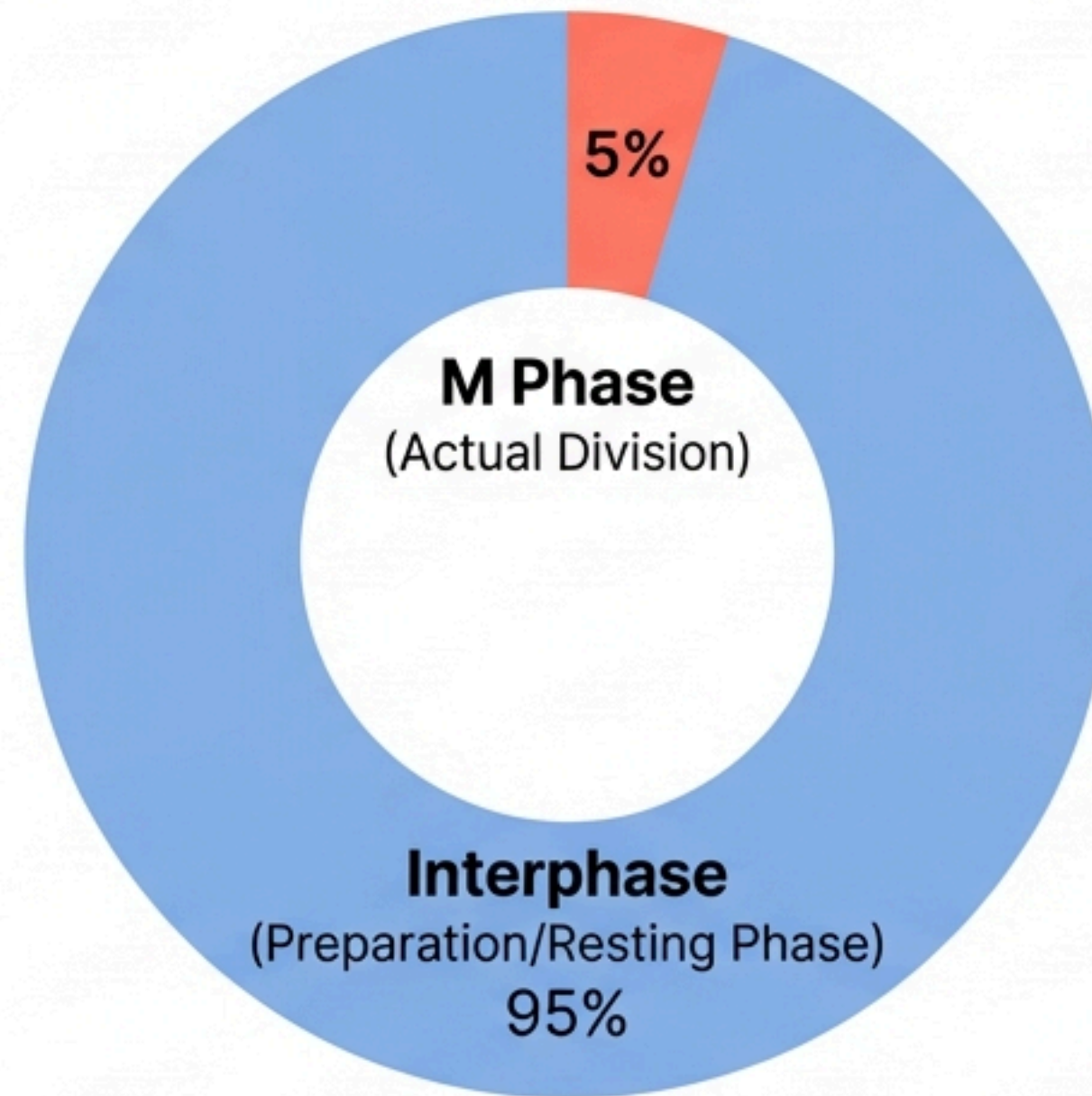
The **Cell Cycle** is the genetically controlled sequence of events by which a cell duplicates its genome, synthesizes other constituents, and eventually divides into two daughter cells.



NEET ALERT: PROCESS TIMING

1. Cell Growth (cytoplasmic increase) is a **continuous** process.
2. DNA Synthesis occurs only during **one specific stage**.

The Cell Cycle Clock: Interphase vs. M Phase



Cycle Duration

- Human Cells: ~24 Hours (M Phase < 1 Hour)
- Yeast: ~90 Minutes

Interphase:

Intense metabolic activity.
Orderly cell growth + DNA replication.

M Phase:

Starts with Nuclear Division (**Karyokinesis**) →
Ends with Cytoplasmic Division (**Cytokinesis**).

Gap 1 (G1) and the Quiescent Stage (G0)



- Cell is metabolically active.
- Continuously grows.
- **DNA does NOT replicate.**

- **Context:** For cells that do not divide (e.g., **Heart Cells**) or divide only for repair.
- **Mechanism:** Cells exit G1 to enter G0.

NEET ALERT: Cells in G0 are **metabolically active** but do **not proliferate** unless called upon. They are suspended, not dead.

S Phase (Synthesis): The Mathematics of Replication

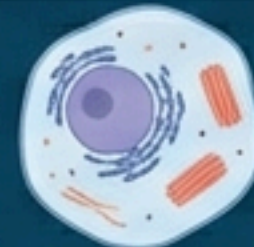


The Golden Rule Table

Parameter	Change
DNA Content	Doubles ($2C \rightarrow 4C$)
Chromosome Number	Remains Same ($2n \rightarrow 2n$)

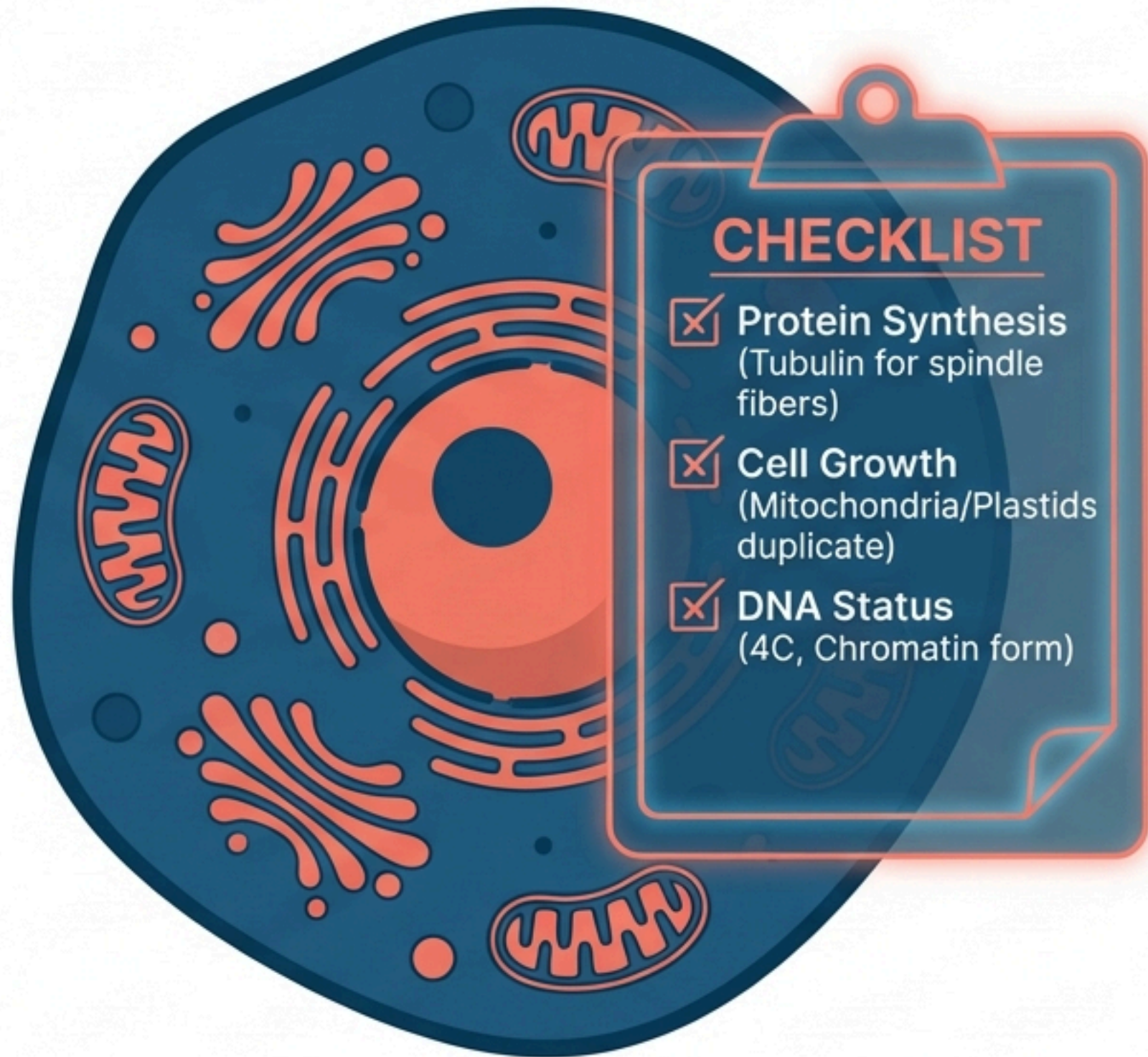


Nucleus: Site of DNA Replication.



Cytoplasm: Site of **Centriole Duplication** (Animal Cells).

Gap 2 (G2): The Final Inspection



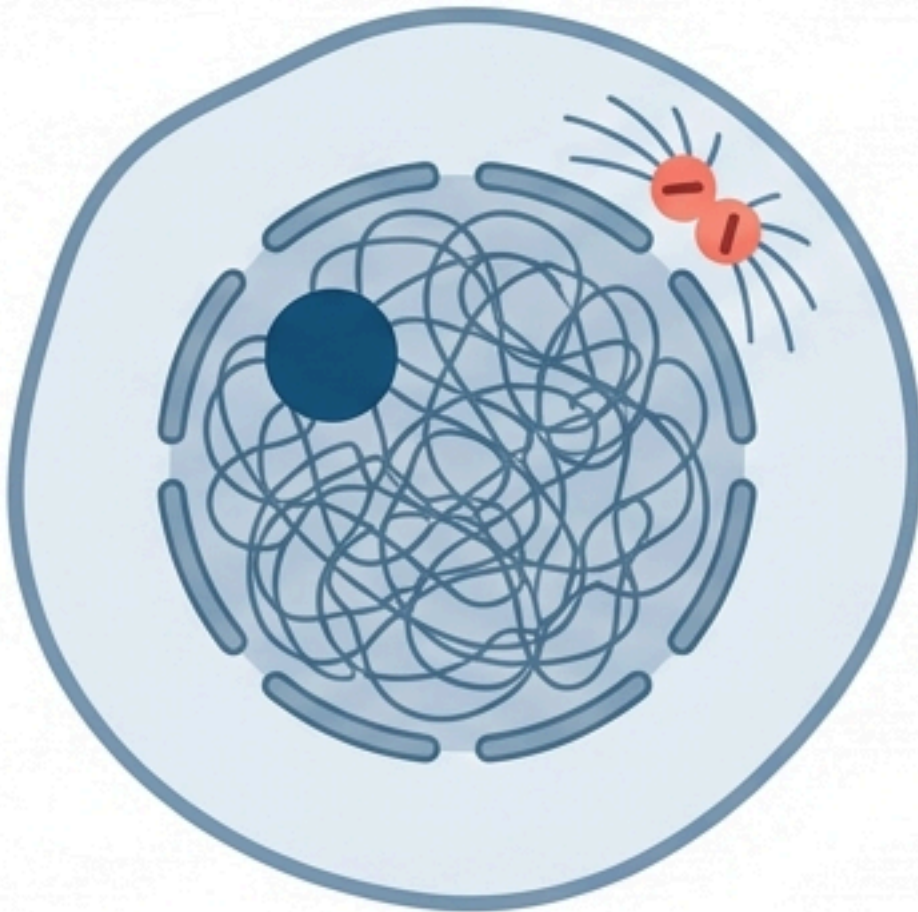
The cell enters a final period of metabolic activity and growth. While DNA replication is finished, the cell must now synthesize the specific proteins required for the mitotic apparatus.

Proceed to M Phase

M Phase - Prophase

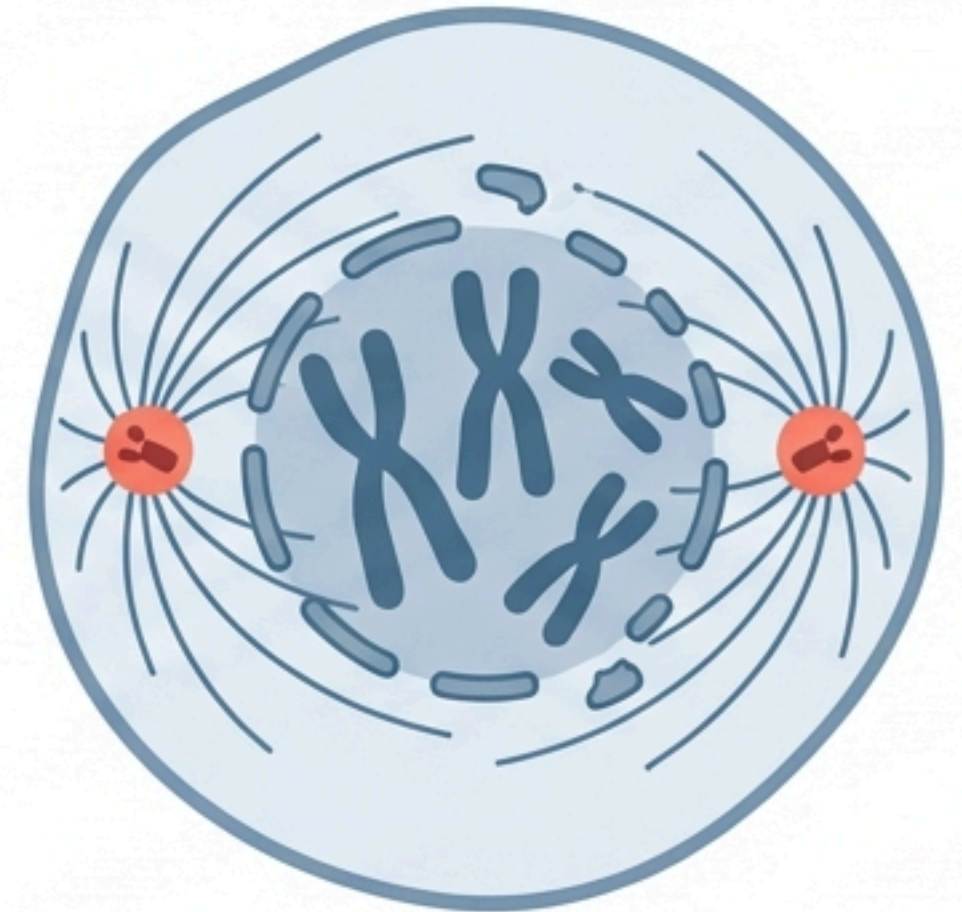
Condensation and Disappearance

Early Prophase



- Chromosomal material untangles and condenses.
- Centrosomes move to opposite poles.
- Formation of Mitotic Apparatus (Spindle fibers + Asters).

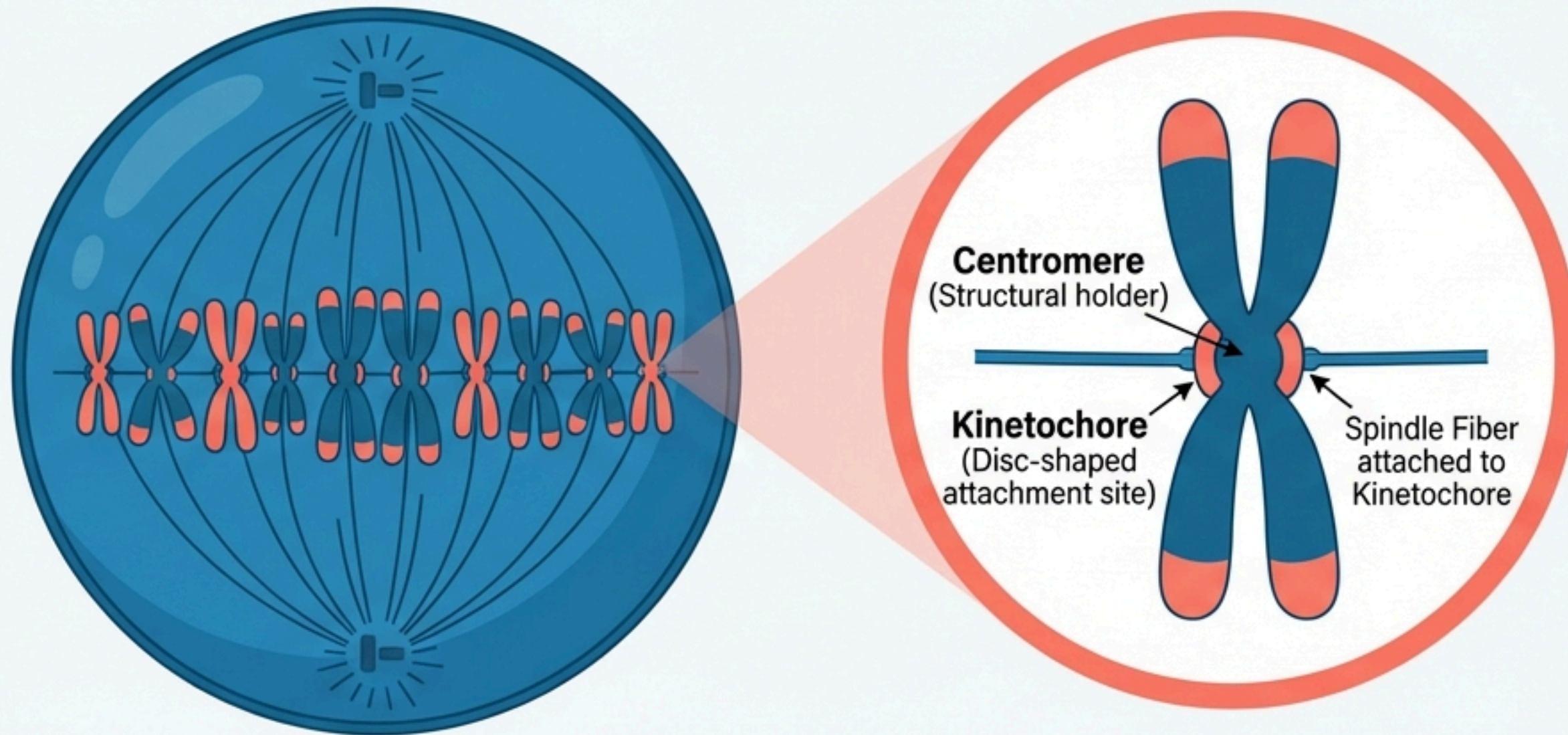
Late Prophase



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The Vanishing Act: By the end of prophase, the cell does **NOT** show: Golgi complexes, ER, Nucleolus, or Nuclear Envelope.

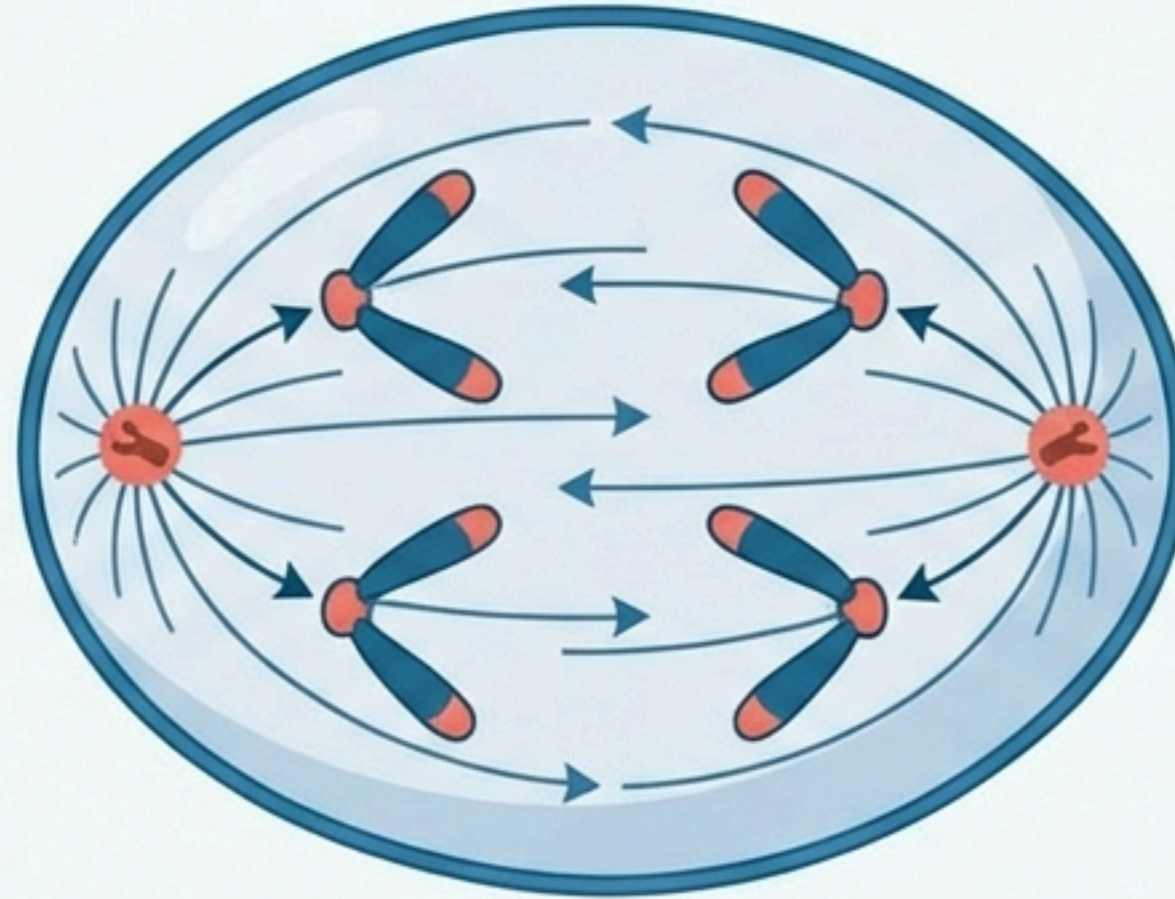
Metaphase: Alignment and Morphology



This is the stage where **morphology of chromosomes** is most easily studied.
Metaphase Plate: The plane of alignment at the equator.

Spindle fibers from opposite poles attach to kinetochores of sister chromatids.

Anaphase: The Great Split



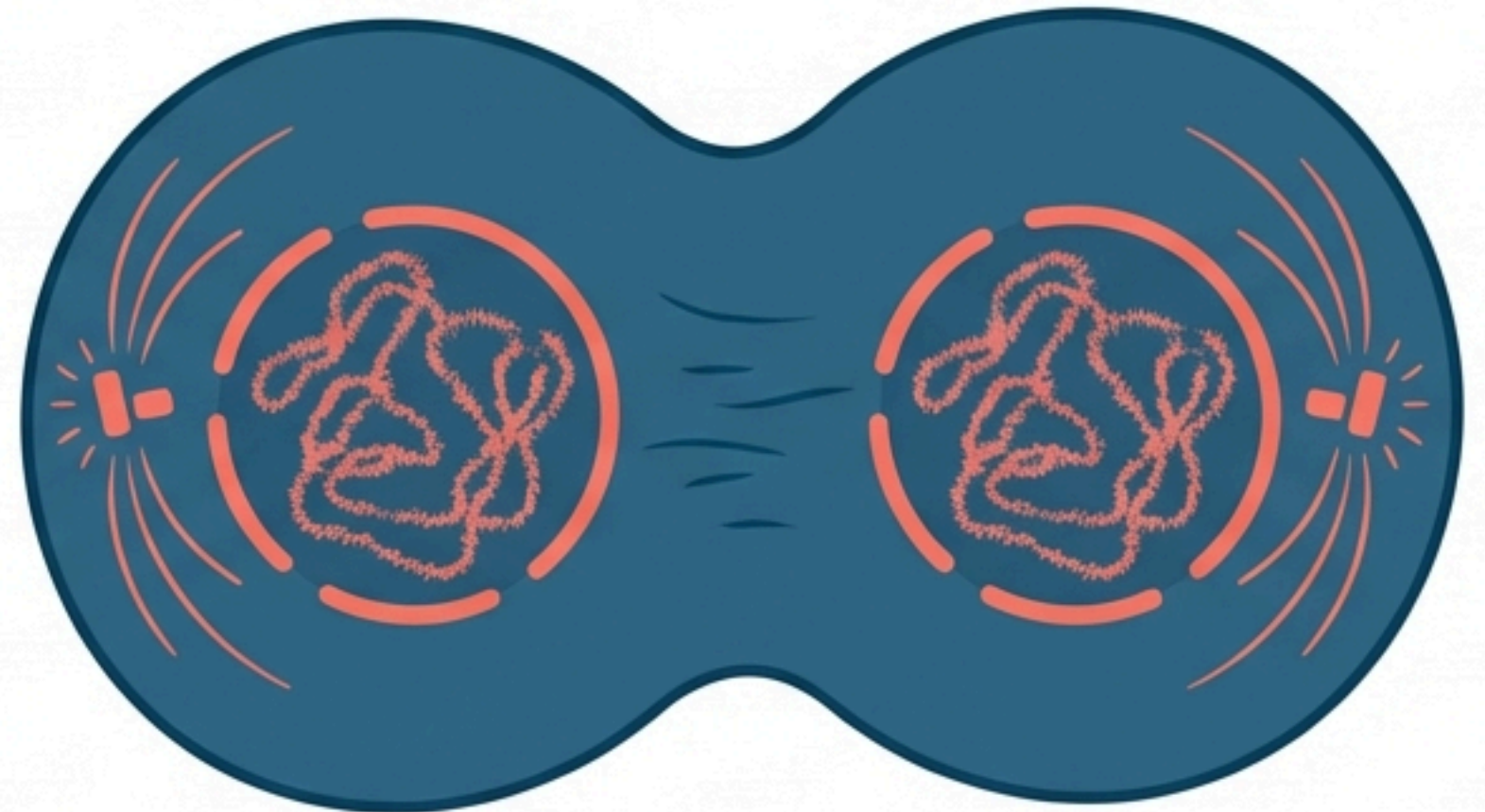
1. Centromeres split.
2. Chromatids separate.
3. Chromatids (now daughter chromosomes) move to opposite poles.

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Leading Edge: The **Centromere** directs toward the pole.
Trailing Behind: The **Arms** of the chromosome.

Telophase: The Reversal

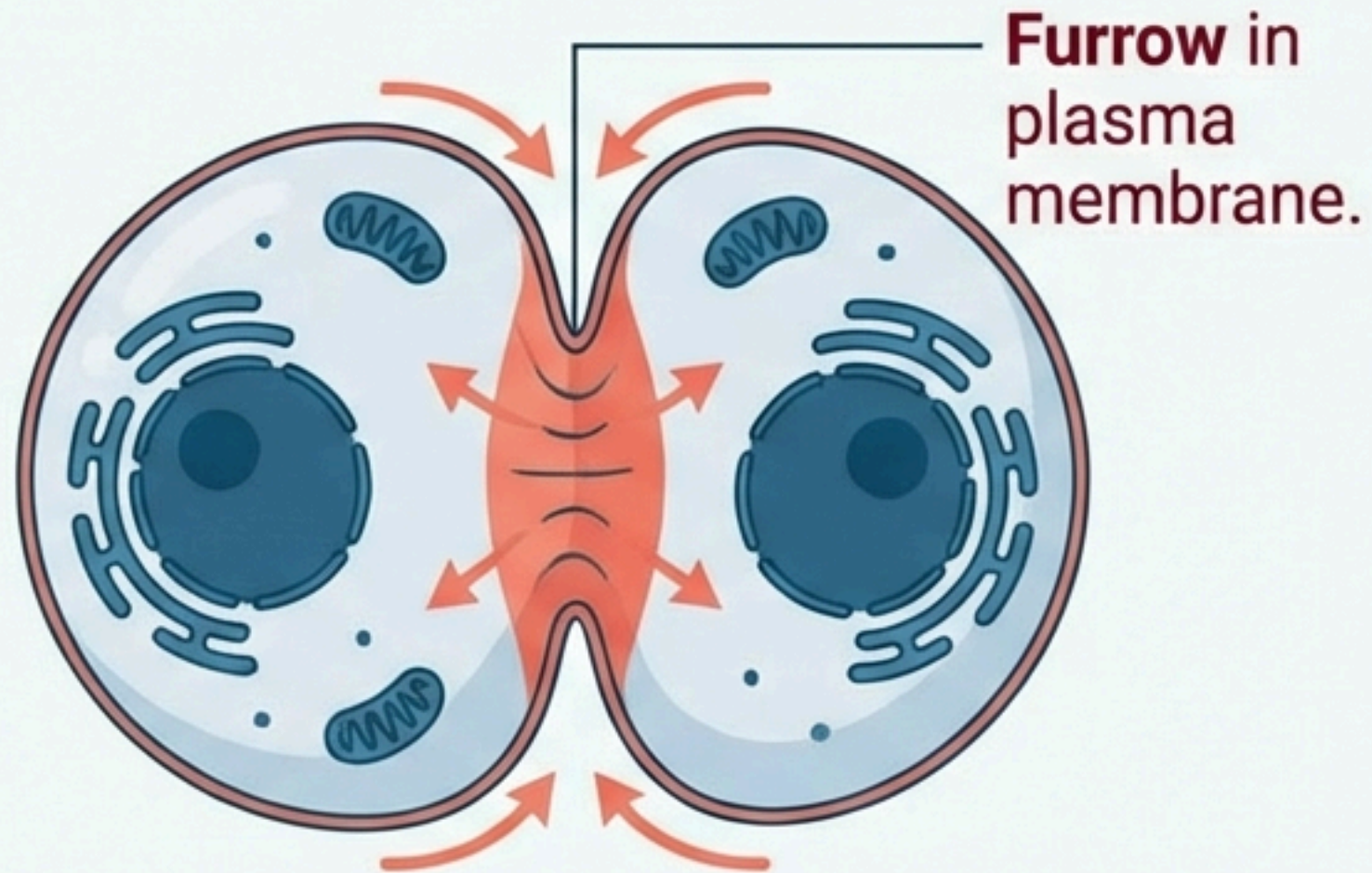
- Reverse Prophase
 - Decondensation: Chromosomes lose individuality → turn back into chromatin.
 - Reformation: Nuclear envelope develops around clusters.
 - Reappearance: Nucleolus, Golgi complex, and ER reform.



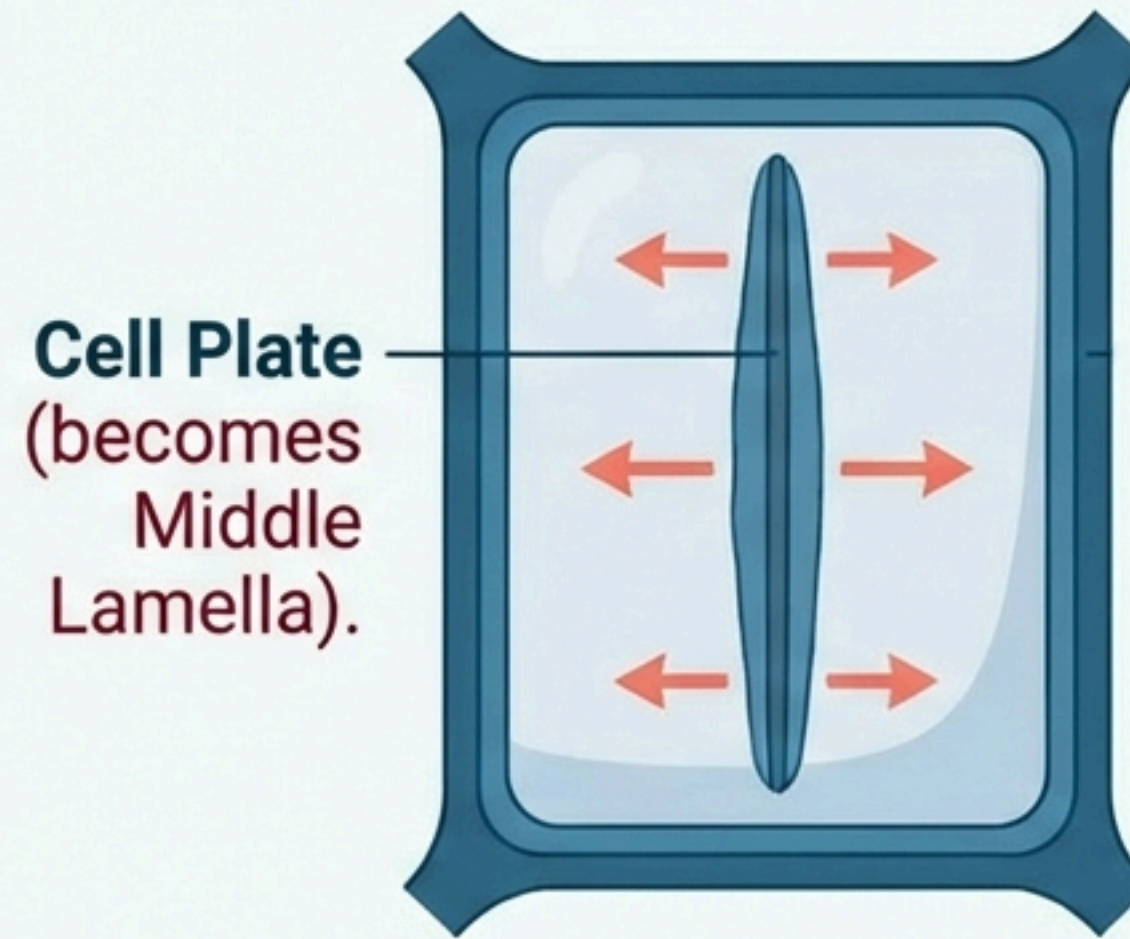
Karyokinesis is complete. Two daughter nuclei are formed.

Cytokinesis: The Physical Division

ANIMAL CELL



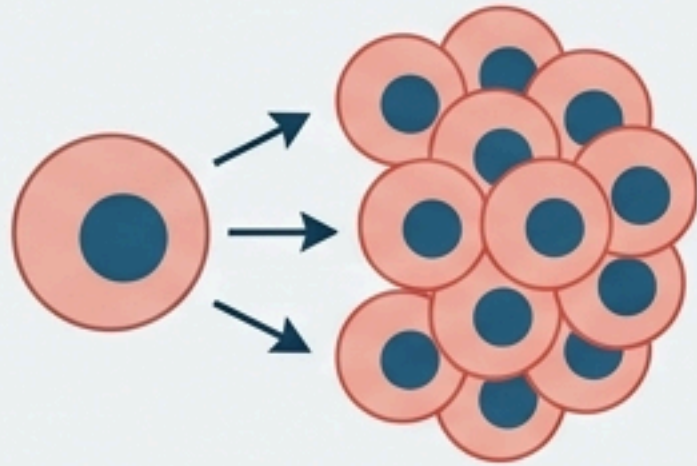
PLANT CELL



**NEET
Alert**

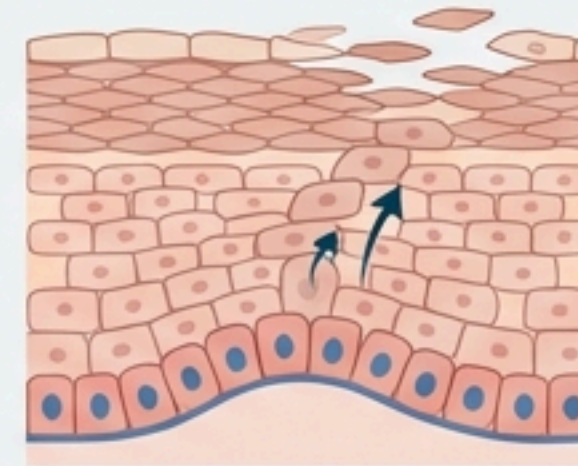
Exception: If Karyokinesis is not followed by Cytokinesis, a multinucleate condition (**Syncytium**) arises. Example: **Liquid Endosperm in Coconut.**

The Significance of Mitosis



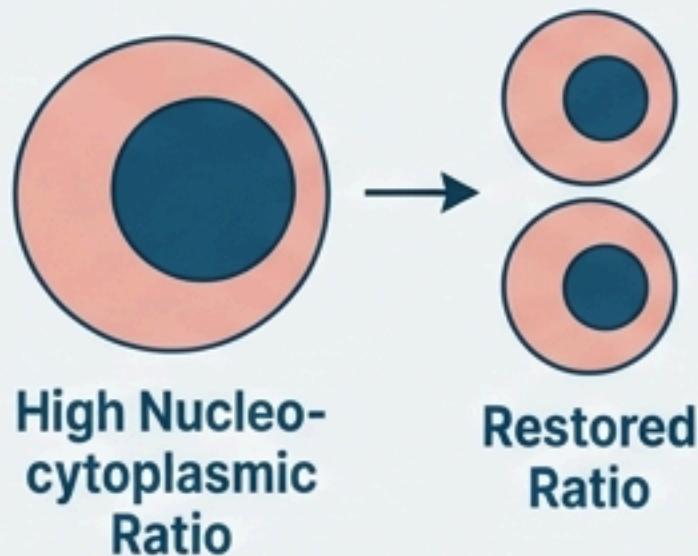
Growth

Essential for multicellular organism growth.



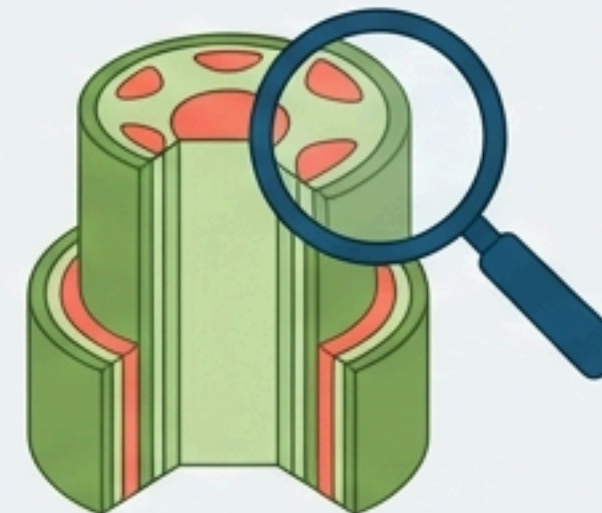
Repair

Constant replacement of Epidermis, Gut Lining, and Blood Cells.



Balance

Restores the **Nucleo-cytoplasmic Ratio** (cell division balances cytoplasmic growth).



Plants

Meristematic tissues (Apical/Lateral Cambium) enable continuous growth.

Haploid Mitosis: Occurs in some lower plants and social insects (e.g., **male honey bees/drones**).

Summary Cheat Sheet: Phase by Phase

Phase	Key Event	DNA Content (Start=2C)	Chromosome # (Start=2n)
G1	Metabolism & Growth	2C	2n
S Phase	DNA Replication	4C	2n
G2	Protein Prep	4C	2n
M Phase (end)	Separation	2C	2n

“M Phase is the most dramatic period involving reorganization of virtually all cell components.”

NEET-UG Challenge: Set A

Q1: In which phase does the amount of **DNA** per cell **double**?

- (A) G1
- (B) G2
- (C) S
- (D) M

Q2: If a cell has **$2n=16$** chromosomes, how many **chromosomes** are present after S phase?

- (A) 8
- (B) 16
- (C) 32
- (D) 64

Q3: **Heart** cells that do not divide are found in which stage?

- (A) G1
- (B) G2
- (C) S
- (D) G0

Q4: In animal cells, **centriole** duplication occurs in the:

- (A) Nucleus
- (B) Cytoplasm
- (C) Mitochondria
- (D) Nucleolus

Q5: The correct sequence of the cell cycle is:

- (A) G1→G2→S→M
- (B) S→G1→G2→M
- (C) G1→S→G2→M
- (D) M→G1→G2→S

NEET-UG Challenge: Set B

Q6: Morphology of chromosomes is best studied during:

- (A) Prophase
- (B) Metaphase
- (C) Anaphase
- (D) Telophase

Q7: During Anaphase, the centromeres are directed:

- (A) Towards equator
- (B) Towards poles
- (C) Randomly
- (D) Towards cell plate

Q8: Which organelles disappear during Prophase?

- (A) Golgi & ER
- (B) Mitochondria only
- (C) Ribosomes
- (D) Vacuoles

Q9: Liquid endosperm in coconut is an example of:

- (A) Cell plate formation
- (B) Syncytium
- (C) Haploid mitosis
- (D) G0 phase